

National Institute of Technology, Rourkela
Mid Semester Examination, Autumn 2022
Compiler Design (CS3007)

5th semester B.Tech in Computer Science & Engineering

Duration: 2 Hours

Full Marks: 60

Number of pages: 1

- Attempt all questions.
- All parts of a question should be answered together.
- In a grammar the upper case letters represent non-terminals, rest of the symbols are terminals.

Q 1. Design a DFA that will recognize the following assembly language constructs [20]

MOV X,Y; Registers $X, Y \in \{A, B, C, D\}$.

MVI X,IH; Register $X \in \{A, B, C, D\}$ and immediate value I is a two digit hexadecimal number followed by an H .

ADD X,Y; Registers $X, Y \in \{A, B, C, D\}$.

ADI X,IH; Register $X \in \{A, B, C, D\}$ and immediate value I is a two digit hexadecimal number followed by an H .

SUB X,Y; Registers $X, Y \in \{A, B, C, D\}$.

SBI X,IH; Register $X \in \{A, B, C, D\}$ and immediate value I is a two digit hexadecimal number followed by an H .

CMP X,Y; Registers $X, Y \in \{A, B, C, D\}$.

CPI X,IH; Register $X \in \{A, B, C, D\}$ and immediate value I is a two digit hexadecimal number followed by an H .

JMP Li; Label Li where L is followed by an integer $i \in \{0, 1, \dots\}$ in decimal.

JEQ Li; Label Li where L is followed by an integer $i \in \{0, 1, \dots\}$ in decimal.

JGT Li; Label Li where L is followed by an integer $i \in \{0, 1, \dots\}$ in decimal.

JLT Li; Label Li where L is followed by an integer $i \in \{0, 1, \dots\}$ in decimal.

Li : ; Label Li where L is followed by an integer $i \in \{0, 1, \dots\}$ and a colon (:).

A semicolon (;) is for end-of-line comments.

Q 2. Consider a grammar G with starting symbol S .

The rule set of G is $\{S \rightarrow iCtSS' | a, S' \rightarrow eS | \epsilon, C \rightarrow b\}$.

(a) Verify whether $G \in LL(1)$. [6]

(b) Find the value of k for which $G \in LL(k)$. [4]

(c) Construct the $LL(k)$ parsing table for G if there exists a k for which $G \in LL(k)$. [6]

(d) Give a trace for derivation of *ibtibteibta* using the $LL(k)$ parsing table for G . [4]

Q 3. Given a grammar G with rules $S \rightarrow 1S0 | 0S1 | 10$.

(a) Is G an $LR(0)$ grammar? Verify. [6]

(b) Is G an $SLR(1)$ grammar? Verify. [4]

(c) Is G a $CLR(1)$ grammar? Verify. [6]

(d) Is G an $LALR(1)$ grammar? Verify. [4]

*****All the best*****